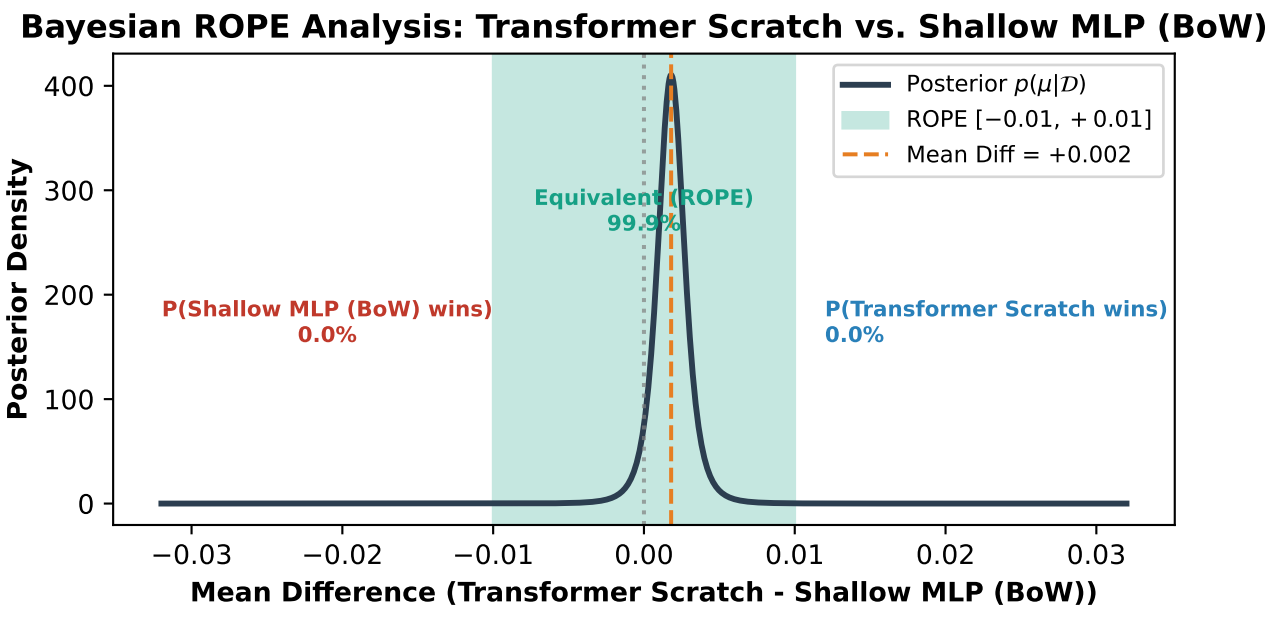
Bayesian ROPE Analysis: Transformer Scratch vs. Shallow MLP (BoW)
Posterior Density
400
300
200
100
0
−0.03
−0.02
−0.01
0.00
0.01
0.02
0.03
Mean Difference (Transformer Scratch - Shallow MLP (BoW))
Posterior $p(\mu|\mathcal{D})$
ROPE [−0.01, +0.01]
Mean Diff = +0.002
Equivalent (ROPE)
99.9%
P(Shallow MLP (BoW) wins)
0.0%
P(Transformer Scratch wins)
0.0%